

Air University



Digital Logic Design (Lab)

Year 2025

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Experiment No. 06

Title: Basic Subtractor Circuit

Objectives:

- To understand the basics of a simple subtractor circuit.
- To design and build a basic subtractor circuit using logic gates.
- To test the circuit's functionality with a truth table and a simulation in Proteus.

Equipment and Components:

- Proteus Software (for simulation)
- Logic Trainer Kit (for practical implementation)
- Logic Gate ICs (7408 - AND, 7486 - XOR, 7404 - NOT)
- Breadboard
- Power Supply (5V)
- Connecting Wires
- Multimeter (optional)

Introduction:

A key element in digital electronics is the basic subtractor circuit, designed to perform binary subtraction between two input bits, A and B. It produces two outputs: the Difference (D) and the Borrow (B_out). This circuit is essential in arithmetic logic units (ALUs) where subtraction operations are executed in digital systems. The basic subtractor serves as a building block for more complex circuits, such as the full subtractor, which can manage borrow inputs from previous stages. Understanding how this circuit is designed and operates is fundamental for developing more advanced digital systems.

This experiment focuses on designing and implementing a basic subtractor circuit using logic gates, specifically XOR, AND, and NOT gates. By verifying its functionality through a truth table and a Proteus simulation, we can confirm that the circuit successfully performs binary subtraction.

Procedure:

1. Design and Simulation:

- a. **Design:** Use one XOR gate, one AND gate, and one NOT gate.
- b. **Implementation:** Connect inputs A and B to the XOR gate to calculate the difference (D). Apply a NOT gate to A to get $\bar{A}$, then connect $\bar{A}$ and B to the AND gate to determine the borrow (B_out).
- c. **Verification:** Simulate the circuit in Proteus and compare the results with the truth table.

2. Implementation on Logic Trainer Kit:

Setup:

- Position the ICs (7486 for XOR, 7408 for AND, 7404 for NOT) onto the Logic Trainer Kit.
- Connect VCC and ground to the appropriate pins on the IC Trainer Kit.

Circuit Connections:

- Attach inputs A and B to the input switches available on the IC Trainer Kit.
- Link the outputs to the output LEDs on the kit.

3. Testing:

- Examine various input combinations according to the truth table.
- Observe the LED status for each set of inputs.
- Optionally, disconnect the output from the LEDs and use a multimeter to check the corresponding voltage readings.

Logic Expression and Truth Table:

Logic Expressions:

- Difference (D) = $A \oplus B$
- Borrow (B_{out}) = $\bar{A} \cdot B$

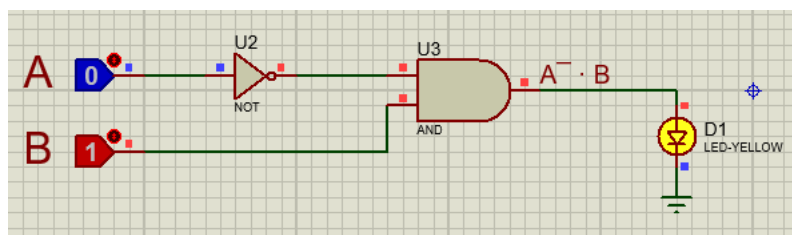
Truth Table:

A	B	D	B _{out}
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

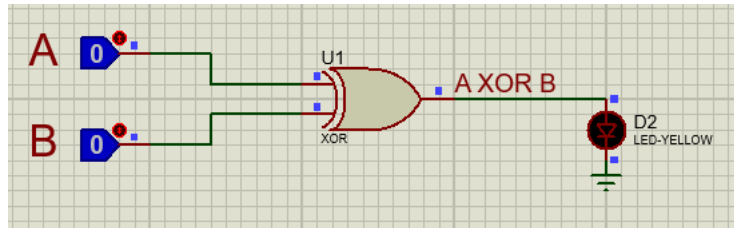
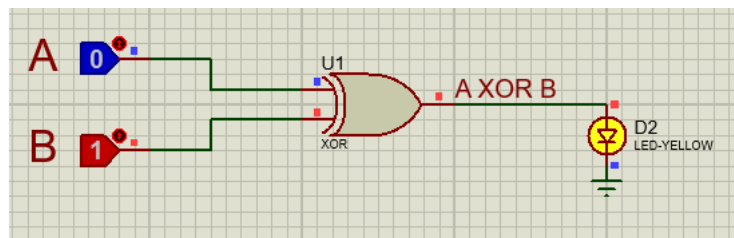
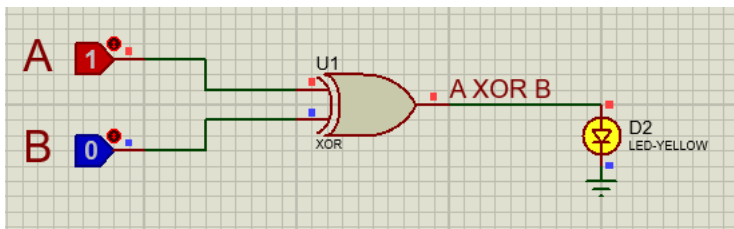
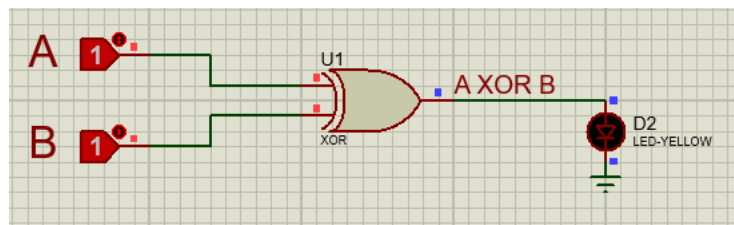
Simulated Results:

i. Verifying: Borrow (B_{out}) = $\bar{A} \cdot B$

Input: A = 0, B = 1 → Output: B_{out} = 1



For all other input combinations the output will be Low (0).

ii. Verifying: Difference (D) = $A \oplus B$ **Input: $A = 0, B = 0 \rightarrow$ Output: $D = 0$** **Input: $A = 0, B = 1 \rightarrow$ Output: $D = 1$** **Input: $A = 1, B = 0 \rightarrow$ Output: $D = 1$** **Input: $A = 1, B = 1 \rightarrow$ Output: $D = 0$** **Conclusions:**

This experiment successfully demonstrated the design and construction of a basic subtractor circuit using XOR, AND, and NOT gates. The truth table alongside the Proteus simulation confirmed the circuit's correctness. The principles underlying binary subtraction were validated, emphasizing the importance of this type of circuit in digital electronics. However, a limitation of this basic design is its inability to handle a borrow-in bit, restricting its use to simple binary subtraction.

THE END